

When you run this analysis, the app is actually asking two separate questions. The **ANOVA** asks the "Big Picture" question, and the **Multiple Comparisons** asks the "Specific Details" question.

Here is the simple, plain-English translation of both tables and their abbreviations.

Table 1: Two-Way Mixed ANOVA Results

The Logic: This table looks at the overall experiment to see if there are *any* broad, mathematical differences. It checks three things:

1. Did the **Group** (WT vs. TG) cause a difference overall?
2. Did the **Timepoint_Weeks** (time passing) cause a difference overall?
3. Was there an **Interaction**? (e.g., Did the TG mice change *differently* over time compared to the WT mice?)

The Abbreviations:

- **Source:** The factor being tested (Group, Timepoint, or the Interaction between them).
- **SS (Sum of Squares):** A raw mathematical number showing the total amount of variation or "wobble" in your data caused by this source.
- **DF1 & DF2 (Degrees of Freedom):** Numbers based on how many subjects, groups, and timepoints you have. They are just used for the background math.
- **MS (Mean Square):** The average variance (SS divided by DF).
- **F (F-Statistic):** Think of this as the **Signal-to-Noise Ratio**. A high F-value means the "signal" (the real difference between your mice) is much louder than the "noise" (random biological variation).
- **p-unc (Uncorrected P-Value):** The probability that the difference you are seeing happened purely by random chance. **If this is less than 0.05, it is statistically significant.**
- **np2 (Partial Eta-Squared):** The **Effect Size**. The p-value tells you *if* there is a difference; np2 tells you *how big* that difference is.
 - ~0.01 = Small effect
 - ~0.06 = Medium effect
 - 0.14 = Large effect

- **eps / p-GG-corr:** Sometimes repeated-measures data violates a mathematical rule called "sphericity" (basically, the variances between timepoints aren't equal). Pingouin is smart enough to detect this and provides a "corrected" p-value (p-GG-corr) to keep your math safe.

Table 2: Multiple Comparisons (Post-Hoc)

The Logic: If the ANOVA says "Yes, there is a difference somewhere!", it doesn't tell you exactly *where* it is. The Post-Hoc table zooms in and compares Group A vs. Group B at every single individual timepoint (e.g., WT vs TG at Week 4, then WT vs TG at Week 8).

The Abbreviations:

- **Timepoint_Weeks:** The specific week being analyzed in this row.
- **A & B:** The two specific groups being compared (e.g., A = WT, B = TG).
- **T (T-Statistic):** Similar to the F-statistic above, but for comparing just two specific groups instead of the whole experiment.
- **dof (Degrees of Freedom):** Based on the number of mice in this specific comparison.
- **p-unc (Uncorrected P-Value):** The raw probability that the difference at this specific week happened by chance. **(Do not use this one!)**
- **p-corr (Corrected P-Value): Use this one!** Because you are running multiple tests (checking Week 4, then Week 8, then Week 12), your chances of getting a "false positive" skyrocket. The p-corr uses a strict mathematical penalty (the Holm method) to adjust the p-value, ensuring your results are scientifically rigorous and publishable. If $p\text{-corr} < 0.05$, that specific week has a significant difference.

Summary for your analysis: Look at the **p-unc** in the ANOVA table to see if your experiment worked overall. If it did, look at the **p-corr** in the Multiple Comparisons table to find exactly which weeks the WT and TG mice diverged.